Effects of Surface Roughness on Aerodynamic Performance of a Wind Turbine Airfoil

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Abstract—**The two-dimensional incompressible Navier-Stokes equations and the SST k-ω turbulence model are used to investigate the aerodynamic performance of a wind turbine airfoil under clean and rough surface conditions.The DU 95-W-180 airfoil is analysized, which is widely used in wind turbines.The numerical simulation of the airfoil under clean surface condition is performed,and the numerical results have a good consistency with the experimental data.The studies in the effects of surface roughness on performance of the airfoil are mainly as follows:computation of the lift coefficient and the drag coefficient of the airfoil under different roughness heights on full surface and different roughness locations;the trends of the lift coefficient and the drag coefficient changing with the roughness heights and roughness locations;the critical value of roughness height and roughness location;the trends of the lift coefficient and the drag coefficient changing with the roughness heights at the critical locations.Furthermore, the unsteady flow field analysis of vortex shedding induced by the surface roughness is performed.**

Keywords-airfoil; roughness; lift coefficient; drag coefficient

I. INTRODUCTION

With the development of science and technology,energy crisis has become a serious problem which we have to face to.As an effective substitute for fossil fuels,solor energy,wind energy and so on are paid close attention.When wind energy is to be concerned,it is very important to study in aerodynamic performance of wind turbine airfoils because it is the basic of wind turbine design.Roughness is one of the important parameters that affect aerodynamic characteristics of wind turbine airfoil.Many investigations have been done to the effects.The full two-dimensional Navier-Stokes algorithm and the SST k-ω turbulence model have been used to predict the performance of the wind turbine airfoil[1].Adapt rough tape near the trailing edge could improve lift and drag coefficient of airfoil[2].The current Navier–Stokes algorithm is capable of predictting the unsteady velocity fluctuations and vortex shedding induced by the large-scale roughness[3].Despite some previous experimental and numerical work where surface roughness is involved,the information that has been obtained on this subject still remains far from complete[4-5].

The purpose of this study is to provide more and more understanding for effects of surface roughness on the performance of the wind turbine airfoil.It is very useful for designers to predict loads and energy losses during wind turbines operating under dust conditions.So the viscous flow around DU 95-W-180 airfoil,special airfoil for wind turbines which is often located at the tip of blade and is of the 18% thickness to chord length ratio,is investigated base on the two-dimensional incompressible Navier-Stokes equations and SST k-ω turbulence model in the study described in this paper.

II. NUMERICAL MODELING

A. Governing Equations

The governing equations are Reynolds averaged Navier-Stokes equations for two-dimensional, viscous, incompressible flow and continuous equation.They are described as follows:

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{1}$$

$$\frac{\partial(\rho\overline{u_i})}{\partial t}+\frac{\partial(\rho\overline{u_i}\overline{u_j})}{\partial x_j}=-\frac{\partial\overline{p}}{\partial x_i}+\frac{\partial}{\partial x_j}(\mu\frac{\partial\overline{u_i}}{\partial x_j}-\rho\overline{u_i'u_j'}) \tag{2}$$

B. SST k-ω Turbulence Model

The shear-stress transport (SST) k-ω model was developed by Menter to effectively blend the robust and accurate formulation of the k-ω model in the near-wall region with the free-stream independence of the k-ω model in the main field[6]. The SST k-ω model is described as follows:

$$\frac{\partial(\rho k)}{\partial t}+\frac{\partial}{\partial x_i}(\rho k u_i)=\frac{\partial}{\partial x_j}(\Gamma_k\frac{\partial k}{\partial x_j})+\widetilde{G_k}-Y_k+S_k \tag{3}$$

$$\frac{\partial(\rho\omega)}{\partial t}+\frac{\partial}{\partial x_i}(\rho\omega u_i)=\frac{\partial}{\partial x_j}(\Gamma_\omega\frac{\partial\omega}{\partial x_j})+G_\omega-Y_\omega+D_\omega+S_\omega \tag{4}$$

where k represents the turbulence kinetic energy;ω represents the specific dissipation rate;G_k represents the generation of turbulence kinetic energy due to mean velocity gradients;G_ω represents the generation of ω;Γ_k and Γ_ω represent

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the effective diffusivity of k and ω respectively;Y_k and Y_ω represent the dissipation of k and ω due to turbulence;D_ω represents the cross-diffusion term,S_k and S_ω are user-defined source terms.

C. Modeling

DU 95-W-180 airfoil(shown in figure1) is a tip airfoil with 18% relative thickness.Its tests were performed in the Low-speed Low-turbulence wind tunnel of the Faculty of Aerospace Engineering of Delft University. The free-stream turbulence level in a 2.6m long,1.25m high and 1.80m wide octagonal test section is 0.07% at 75m/s,with Reynolds number $3x10^6$, using 0.6m chord models[7].In this study,the chord length of the model is 0.6m,and the longitudinal section of the wind tunnel test section is selected to be the computational zone.And in order to improve accuracy of simulation,the section is extended to 7.8m(shown in figure 2).Hybrid structured-unstructured grids are used to the simulation.The section of the computational zone near the airfoil is C-type structured grids,and the other section is unstructured grids.The number of the grid nodes is about 210,000.The height of the first grid around the airfoil is 0.005mm.

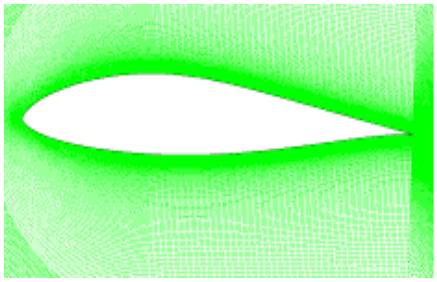


Figure 1. DU 95-W-180 airfoil

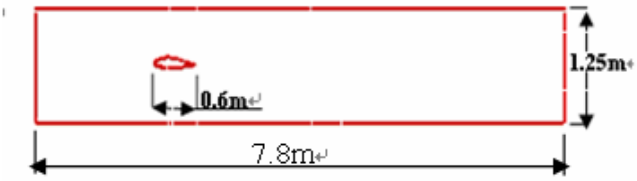


Figure 2. The computational zone

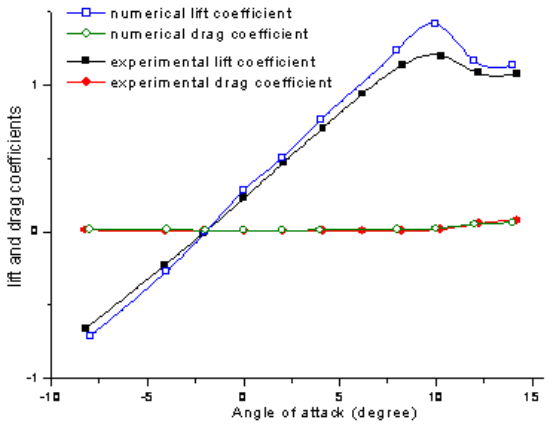


Figure 3. Experimental and simulational data of lift coefficient and drag coefficient

III. Numerical Results and Discussions

A. Numerical Simulation under Clean Surface Condition

Numerical simulation under clean surface condition is performed with the Reynolds number $3x10^6$. The trends of the lift coefficient and the drag coefficient changing with the angle of attack are obtained,and the results are compared with the data of experiments.(shown in figure 3).

Obviously the lift coefficient and the drag coefficient of numerical calculation are consistent with the results of experiments perfectly.That is to say, the two-dimensional Navier-Stokes equations and the SST k-ω turbulence model are available for predicting the performance of the wind turbine airfoil.

B. Numerical Simulation under Different Surface Roughness Height

Numerical simulation under different surface roughness height is performed when the roughness heights are 0.03mm, 0.08mm,0.1mm,0.2mm,0.3mm,0.5mm,0.6mm,0.7mm,1mm,1.5 mm and 2mm respectively.The simulation is performed on the hypothesis that the roughness is distributed uniformly on full surface of the airfoil.The angle of attack is set to 2 degrees to study the effects on laminar-turbulent transition on the roughness surface of the airfoil.The trends of the lift coefficient and the drag coefficient changing with roughness height are obtained (see figure 4).And the curves of pressure coefficient are also gained at different roughness heights(see figure 5).

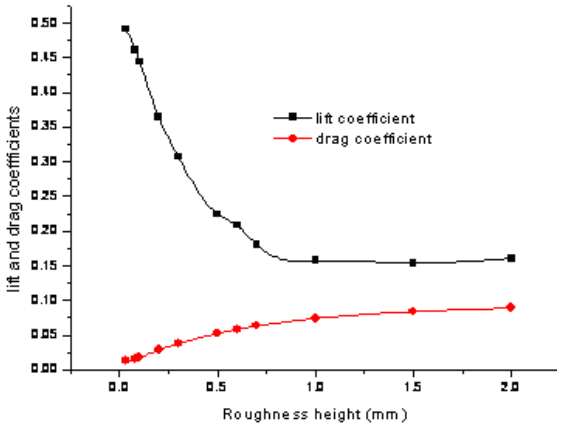


Figure 4. Lift coefficient and drag coefficient with roughness height

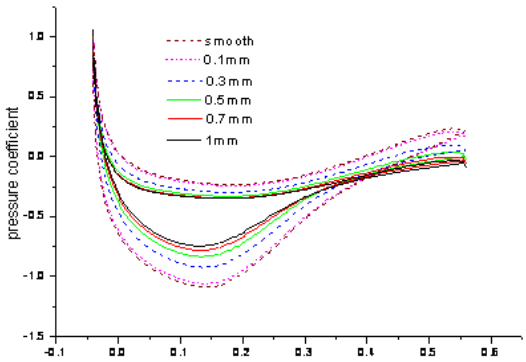


Figure 5. Pressure coefficient at different roughness height

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From the figure 4 and figure 5, the critical roughness height is 0.5mm.The lift coefficient curve decreases very rapidly when the roughness height is less than 0.5mm,and then the trend slows down.However, the lift coefficient curve decreases very slowly when the roughness height is more than 0.7mm.The same trend of the change also happens to the drag coefficient curve. The drag coefficient increases rapidly when the roughness height is less than 0.5mm,and then the trend slows down,but increases slowly when the roughness height is more than 0.7mm.It is why the surface roughness plays an important role in transition from laminar flow to turbulent flow.The effect of roughness is more obvious when the roughness height is less than 0.5mm.With the increasing of the roughness height, the laminar boundary layer around the airfoil gradually become turbulent one.The effect of roughness on the performance of the airfoil is no longer strenuous when the roughness height is more than 0.7mm,so the trend slows down.Comparing with the conclusion in reference[1](The roughness height of 0.3mm could be viewed as a critical roughness height for the NACA 63-430 airfoil.),It is found that the critical roughness height of airfoils with different thickness,camber and angle of attack is different.

C. Numerical Simulation at Different Roughness Locations

Numerical simulation at different surface roughness locations is performed on locating the rough tape at the position of 5%,10%,20%,30%,60%,100% chord length from leading edge on both pressure and suction surfaces of the airfoil.The rough tape is 10mm wide,and the roughness height is 0.5mm because it is the critical height of the airfoil. The angle of attack is 2 degrees.The lift and drag coefficients changing with roughness locations are obtained(see figure 6,7).

From the figure,the critical locations are 10% of the chord length on suction surface and 100% of the chord length on pressure surface.At 10% of the chord length,the lift coefficient is the minimum.In an other word,roughness effects are serious than other location.Comparing with the conclusion of reference [1](The roughness location at 25% of the chord length had the most obvious effect for the NACA 63-430 airfoil.),It is found that the critical location of airfoils with different thickness,camber and angle of attack is different.At 100% of the chord length,the lift coefficient is large than the data of clean surface.The drag coefficient is large too.

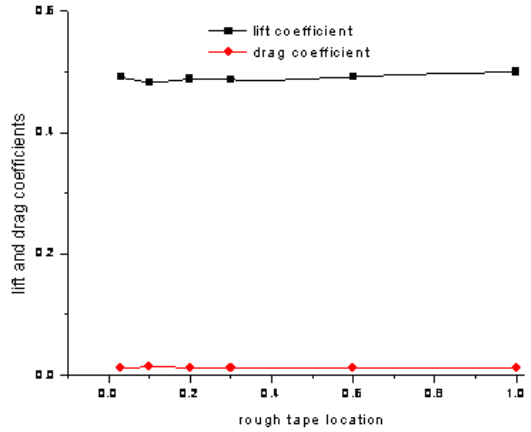


Figure 6. Lift coefficient and drag coefficient with rough tape location on the suction surface

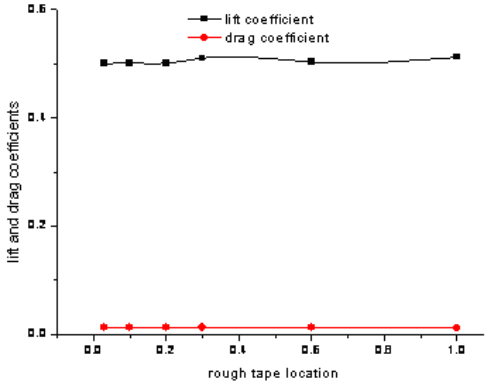


Figure 7. Lift coefficient and drag coefficient with rough tape location on the pressure surface

D. Numerical Simulation at the critical locations

Numerical simulation is performed on the critical locations(10% of the chord length on suction surface and 100% of the chord length on pressure surface).The trends of the lift and drag coefficient changing with roughness height are obtained respectively(see figure 8,9).

At 10% of the chord length on suction surface,as we can see from the figure 8,the lift coefficient decreases when roughness height increases from 0mm to 0.5mm,but,decreases very slowly when roughness height is more than 0.5mm.At the same time,the drag coefficient increases when roughness height increases from 0mm to 0.5mm,but, increases very slowly when roughness height is more than 0.5mm.On the other hand, it is verified that 0.5mm roughness height is critical to the airfoil.

At 100% of the chord length on pressure surface,from the figure 9,the lift coefficient decreases slowly when roughness height increases from 0mm to 0.1mm.However,it increases when roughness height increases from 0.1mm to 0.7mm,and increases very slowly when roughness height is more than 0.7mm.At the same time,the drag coefficient increases when roughness height increases from 0mm to 1.5mm.In an other word,we can improve lift coefficient by the method of increasing adapt roughness at the location near the trailing adge on pressure surface.

In order to study the effect of flow field under roughness surface,the numerical simulation is performed at the 2 degrees angle of attack,and locating a isolated hump(1mm high,2mm wide)at 10% of the chord length on suction surface.The gragh of the velocity cross vorticity is obtained (shown in figure 10).The figure 11 is the local amplification of figure 10.The primary goal of this part is to investigate unsteady flow and vortex shedding past large-scale roughness.

It is well known that there is a laminar flow on the clean surface of airfoil when the angle of attack is 2 degrees.But,as we can see from the figures,the laminar flow has transferred to turbulent flow near the hump,and we can see shedding vortices and flow separation obviously.For the existence of the vortices and the flow separation,the lift coefficient decreases 8%,and the drag coefficient increases 23% of the data of clean surface.On the other hand,we can say that the two-dimensional incompressible Navier-Stokes equations and the SST k-ω

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turbulence model are available to predict the flow separation and snap shedding vortices.

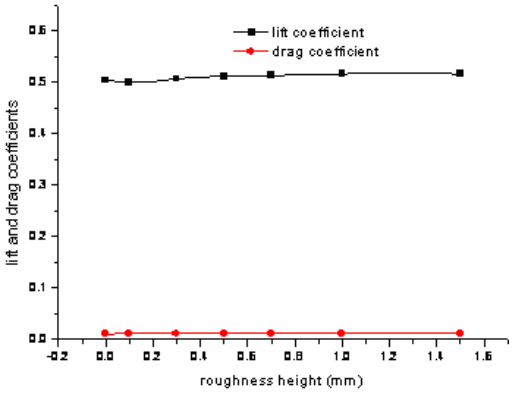


Figure 8. Lift coefficient and drag coefficient with roughness height at the 10% chord length

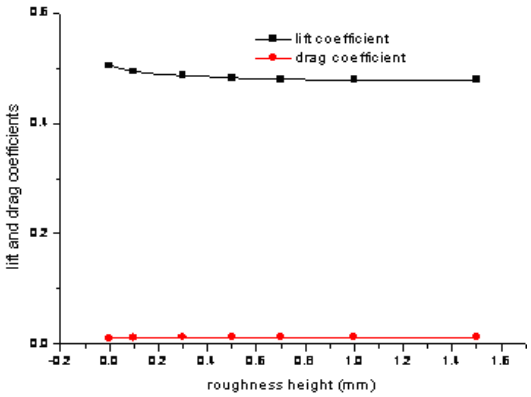


Figure 9. Lift coefficient and drag coefficient with roughness height at the 100% chord length

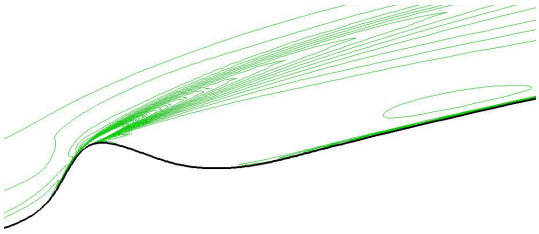


Figure 10. The velocity cross vorticity induced by the isolated hump at the 10% of chord length

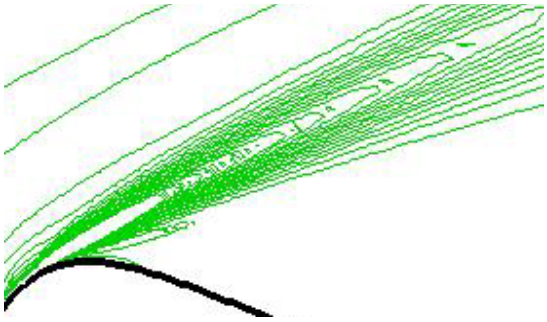


Figure 11. Local amplification of figure 10

IV. CONCLUSIONS

The two-dimensional Navier-Stokes equations and the SST k-ω turbulence model are available to predicting the aerodynamic characteristics of the wind turbine airfoil, the flow separation and the shedding vortices of the flow field.

The critical roughness height is 0.5mm for DU 95-W-180 airfoil. The lift coefficient curve decreases very rapidly when the roughness height is less than 0.5mm,but decreases slowly when the roughness height is more than 0.7mm. The drag coefficient increases rapidly when the roughness height is less than 0.5mm, but increases slowly when the roughness height is more than 0.7mm. The critical roughness height of airfoils with different thickness,camber and angle of attack is different..

The critical locations are 10% of the chord location on suction surface and 100% of the chord location on pressure surface for DU 95-W-180 airfoil.At 10% of the chord location on suction surface,lift coefficient is minimum.But the critical location of airfoils with different thickness,camber and angle of attack is different.At 100% of the chord location on pressure surface, the lift coefficient and the drag coefficient are large than the data of clean surface respectively.

At 10% of the chord location on suction surface, the lift coefficient decreases more rapidly when roughness height increases from 0mm to 0.5mm,but,decreases very slowly when roughness height is more than 0.5mm.At 100% of the chord location on pressure surface, the lift coefficient decreases when the roughness height increases from 0mm to 0.1mm,but increases when roughness height is more than 0.1mm.The drag coefficient increases when roughness height increases from 0mm to 1.5mm. We can improve lift coefficient by the method of increasing adapt roughness at the location near the trailing adge on pressure surface.

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